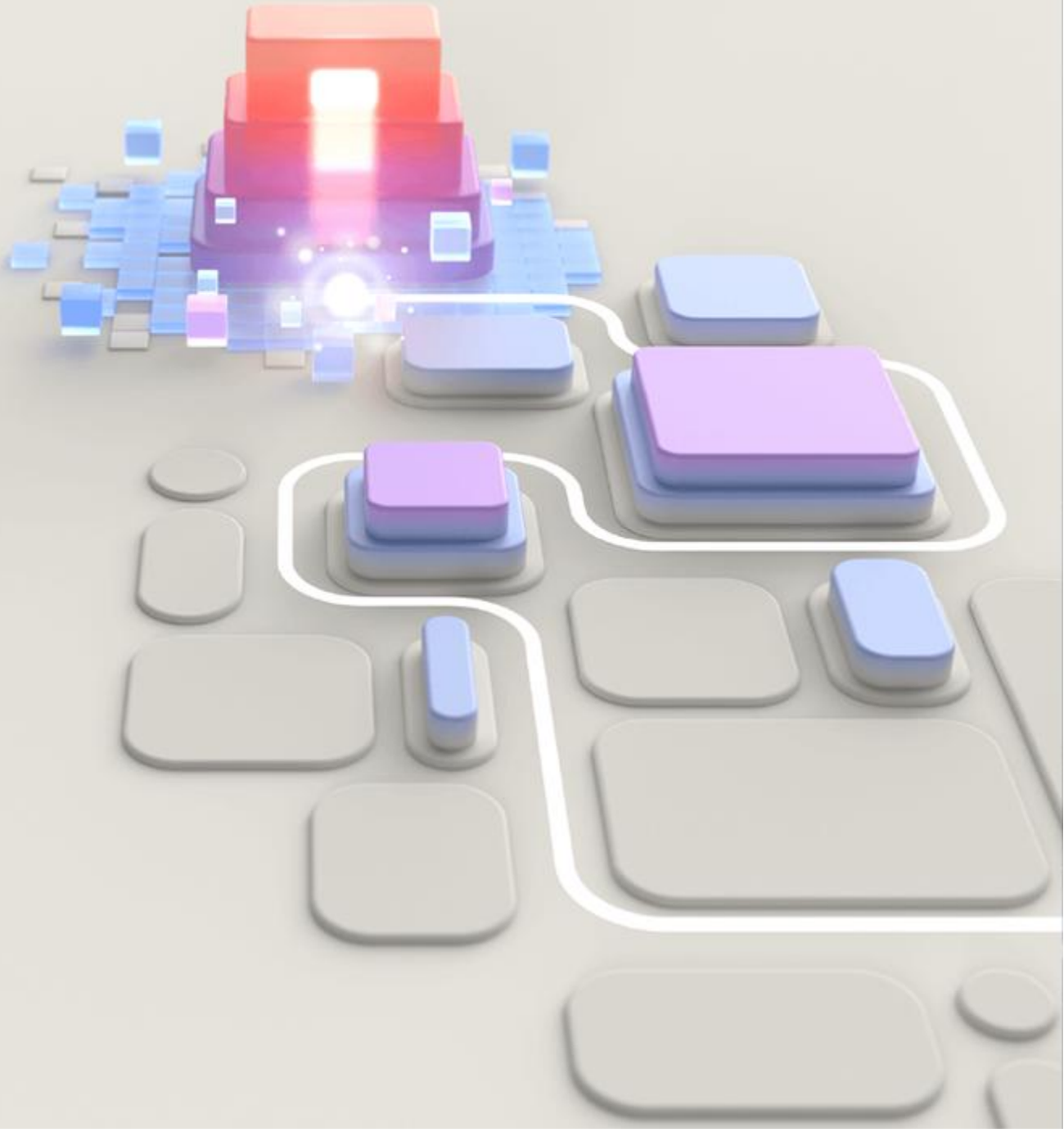
Microsoft

AZ-104

Administer PaaS Compute Options



© Copyright Microsoft Corporation. All rights reserved.

AZ-104 Agenda

01: Administer Identity

02: Administer Governance and Compliance

03: Administer Azure Resources

04: Administer Virtual Networking

05: Administer Intersite Connectivity

06: Administer Network Traffic Management

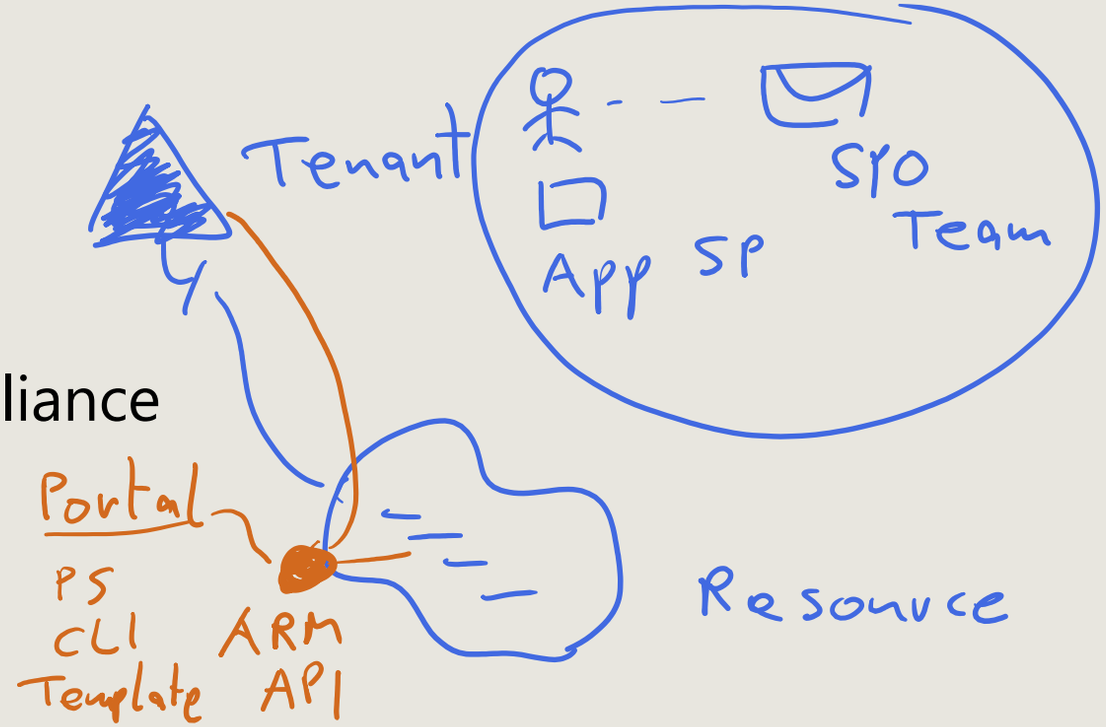
07: Administer Azure Storage

08: Administer Azure Virtual Machines

09: Administer PaaS Compute Options

10: Administer Data Protection

11: Administer Monitoring



Container (Docker, ARO)
K8S

App GW — WAF

App Service

Backup
Site Recovery

LAW
Log Analytics Workspace

Administer PaaS Compute Options Introduction

- Configure Azure App Service Plans
- Configure Azure App Services
- Configure Azure Container Instances
- Lab 09a - Implement Web Apps
- Lab 09b - Implement Azure Container Instances
- Lab 09c – Implement Azure Container Apps

© Copyright Microsoft Corporation. All rights reserved.

Configure Azure App Service Plans

© Copyright Microsoft Corporation. All rights reserved.

Learning Objectives - Configure Azure App Service Plans

- Implement Azure App Service Plans
- Determine App Service Plan Pricing
- Scale Up and Scale Out the App Service Plan
- Configure App Service Plan Scaling
- Demonstration – Configure Azure App Service Plans
- Learning Recap

Exam objectives

Create and configure Azure App Service

- Provision an App Service plan
- Configure scaling for an App Service plan

© Copyright Microsoft Corporation. All rights reserved.

Implement Azure App Service Plans

- Determines performance, price, and features
- Defines a set of compute resources for a web app to run
  - Region where compute resources will be created
  - Number of virtual machine instances
  - Size of virtual machine instances
  - Pricing tier (next slide)
- One or more apps can be configured to run in the same App Service plan

© Copyright Microsoft Corporation. All rights reserved.

Determine App Service Plan Pricing

Selected Features	Free	Shared (dev/test)	Basic (dedicated dev/test)	Standard (production workloads)	Premium (enhanced scale and performance)	Isolated (high-performance, security and isolation)
Web, mobile, or API apps	10	100	Unlimited	Unlimited	Unlimited	Unlimited
Disk space	1 GB	1 GB	10 GB	50 GB	250 GB	1 TB
Auto Scale	–	–	–	Supported	Supported	Supported
Deployment Slots	0	0	0	5	20	20
Max Instances	–	–	Up to 3	Up to 10	Up to 30	Up to 100

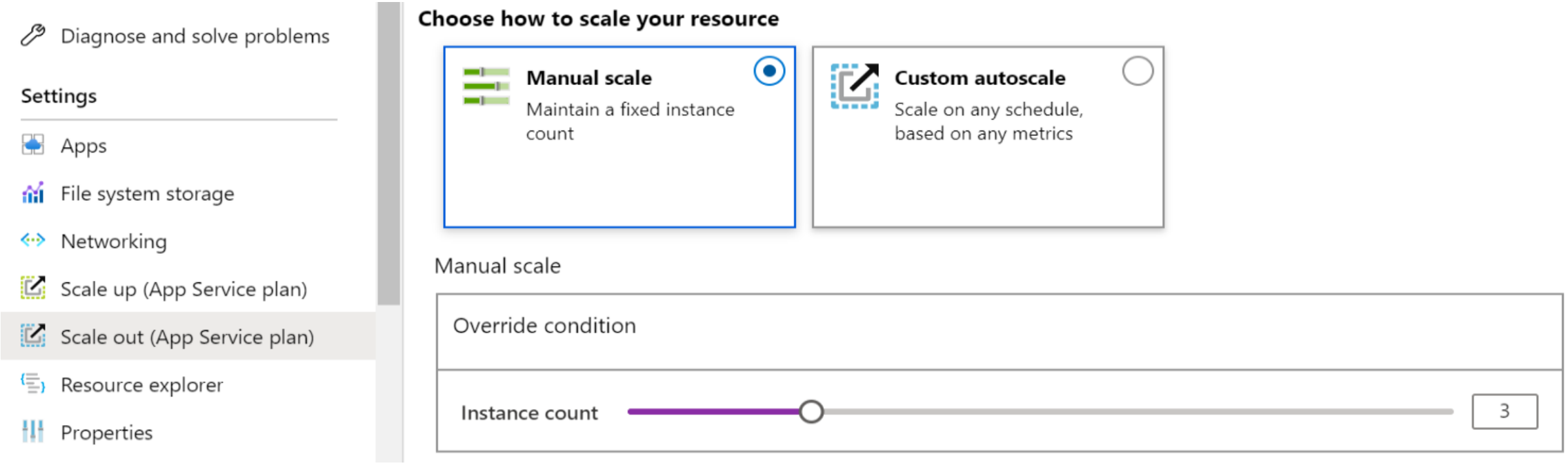
Shared compute
(Free and Shared). Run apps on the same Azure VM as other App Service apps, and the resources cannot scale out

Dedicated compute
(Basic, Standard, Premium). Run apps in the same plan in dedicated Azure VMs

Isolated. Runs apps on dedicated Azure VMs in dedicated Azure virtual networks

© Copyright Microsoft Corporation. All rights reserved.

Scale Up and Scale Out the App Service Plan



Scale up (change the App Service plan):

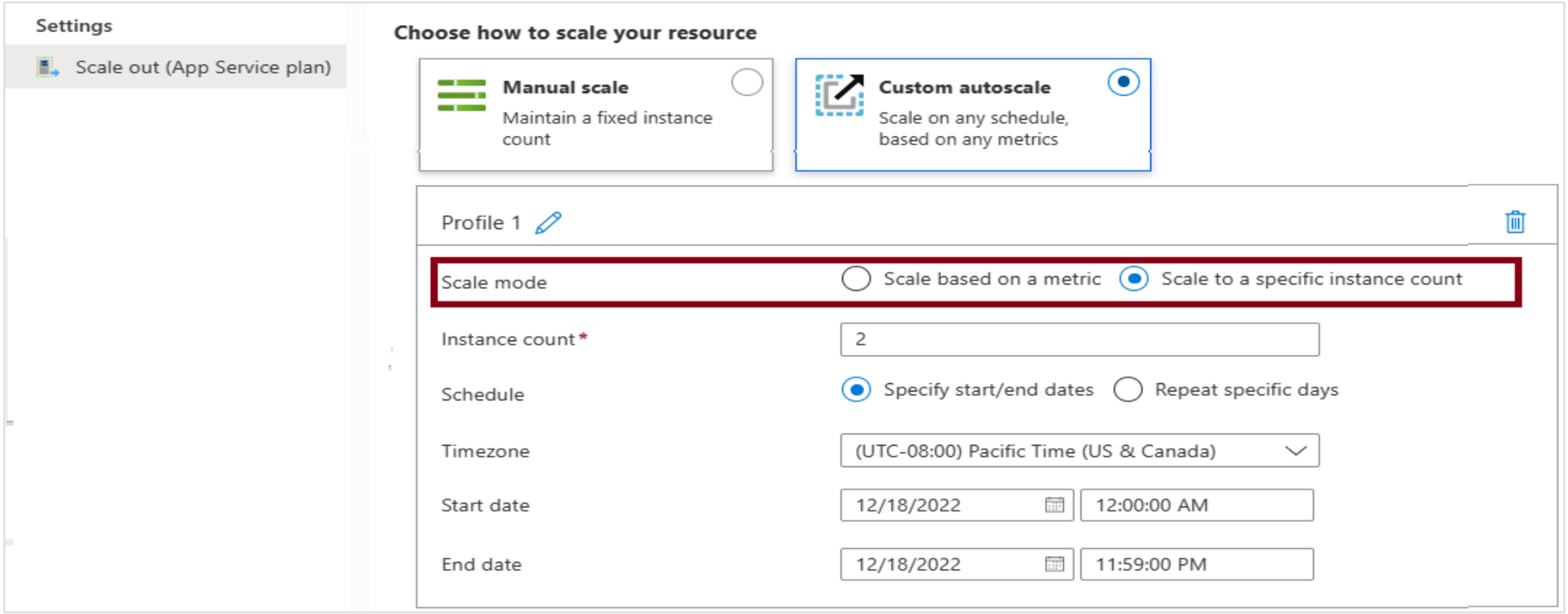
- More hardware (CPU, memory, disk)
- More features (dedicated virtual machines, staging slots, autoscaling)

Scale out (increase the number of VM instances):

- Manual (fixed number of instances)
- Auto scale (based on predefined rules and schedules)

© Copyright Microsoft Corporation. All rights reserved.

Configure App Service Plan Scaling



Adjust available resources based on the current demand

Improves availability and fault tolerance

Scale based on a metric (CPU percentage, memory percentage, HTTP requests)

Scale according to a schedule (weekdays, weekends, times, holidays)

Can implement multiple rules – combine metrics and schedules

Don't forget to scale in

© Copyright Microsoft Corporation. All rights reserved.

Configure Azure App Services

✗ FORTRAN 77

✓ .Net
✓ Python
✓ Java
✓ Node.js
✗ Rust

Platform "Kudu"

Plan A (code)

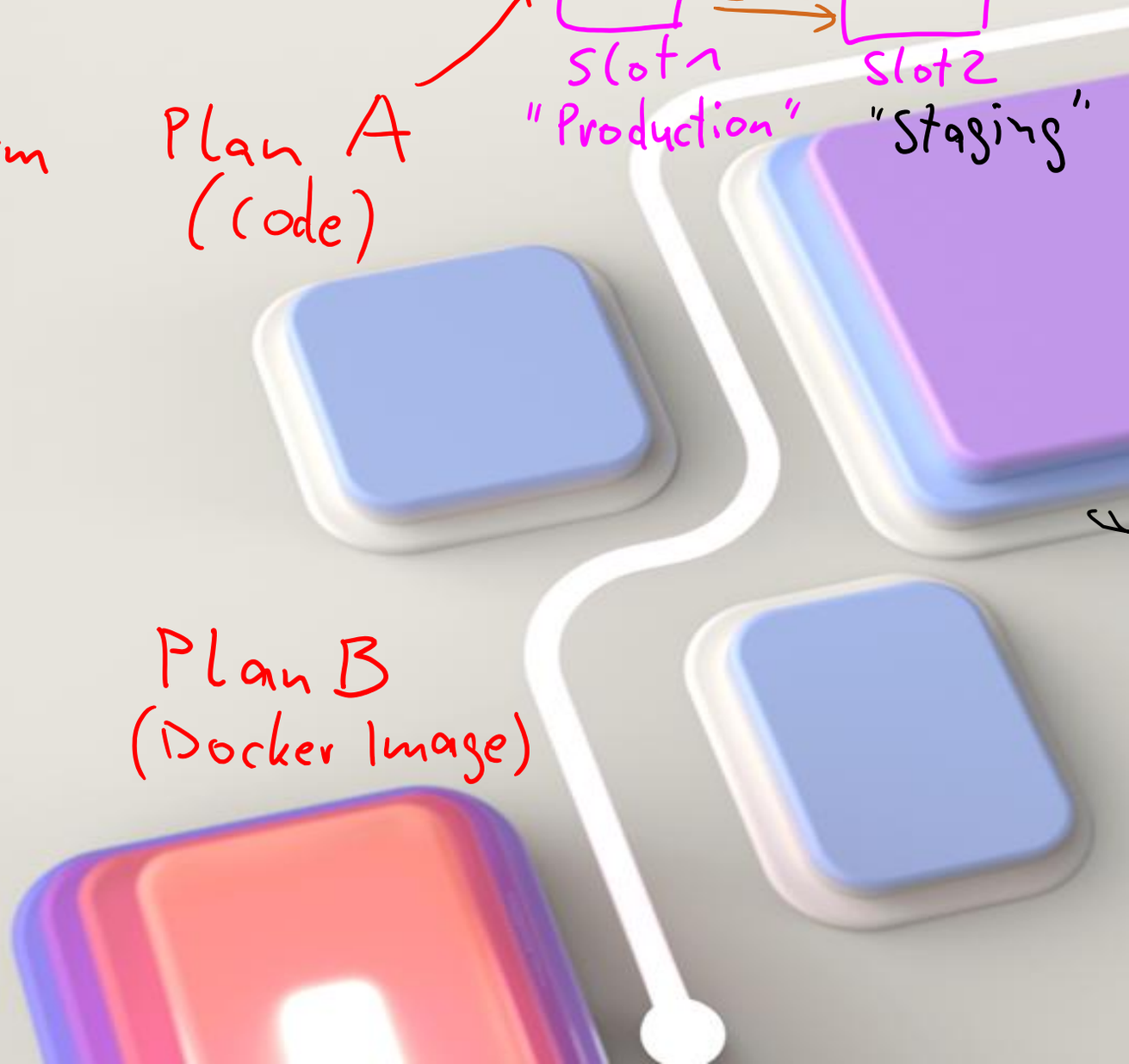
v1 → Slot 1 "Production"
v2 → Slot 2 "Staging"

docker build.

FORTRAN —
Lib —
Windows/Linux —

Image

Plan B (Docker Image)



© Copyright Microsoft Corporation. All rights reserved.

Learning Objectives - Configure Azure App Services

- Implement Azure App Service
- Create an App Service
- Create Deployment Slots
- Add Deployment Slots
- Secure an App Service
- Create Custom Domain Names
- Backup an App Service
- Demonstration – Azure App Services
- Learning Recap

Exam objectives

Create and configure Azure App Service

- Create an App Service
- Configure certificates and TLS
- Map an existing custom DNS name
- Configure backup
- Configure networking settings
- Configure deployment slots

© Copyright Microsoft Corporation. All rights reserved.

Implement Azure App Service

- Includes Web Apps, API Apps, Mobile Apps, and Function Apps
- Fully managed environment enabling high productivity development
- Platform-as-a-service (PaaS) offering for building and deploying highly available cloud apps for web and mobile
- Platform handles infrastructure so developers focus on core web apps and services
- Developer productivity using .NET, .NET Core, Java, Python and a host of others
- Provides enterprise-grade security and compliance

QUICKSTART

ASP.NET

Python

Node.js

PHP

WordPress

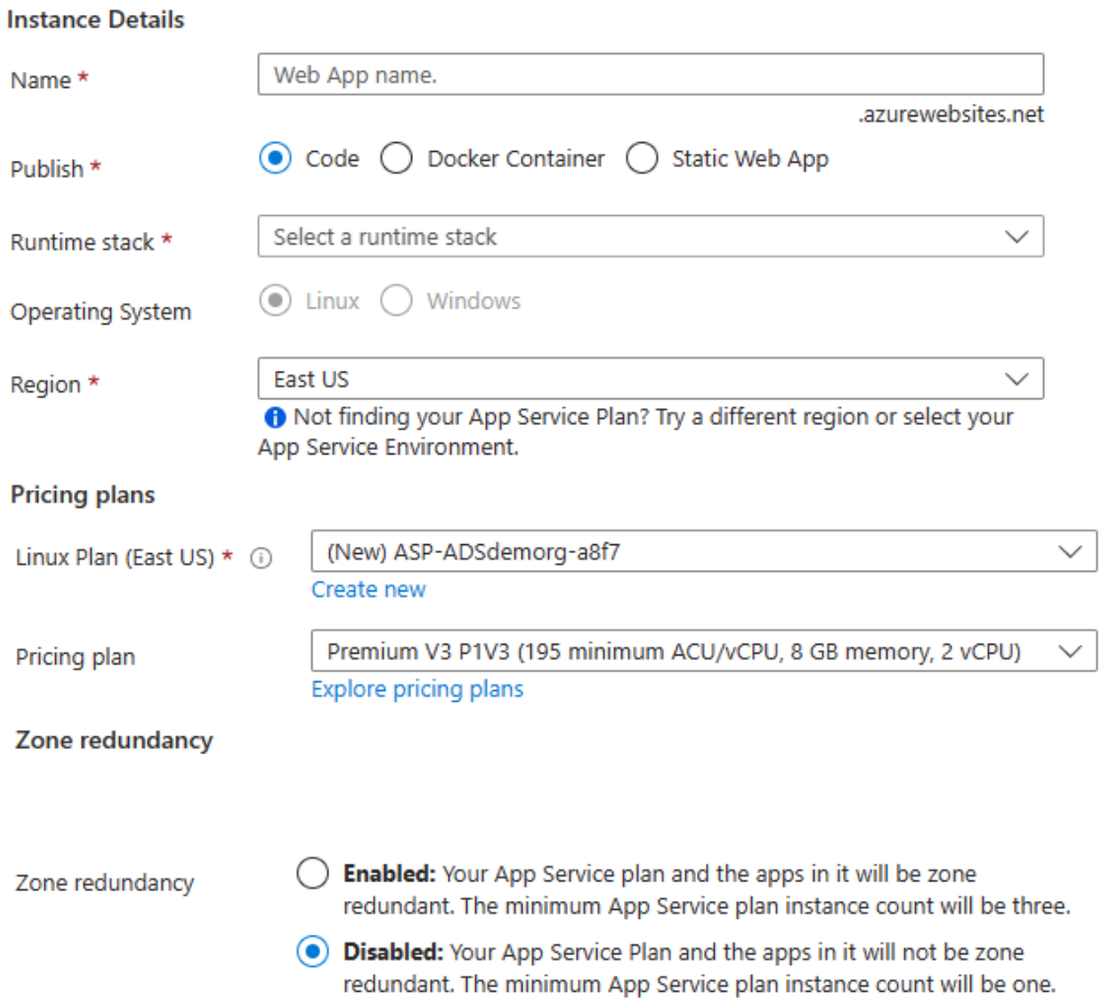
Custom container

Java

© Copyright Microsoft Corporation. All rights reserved.

Create an App Service

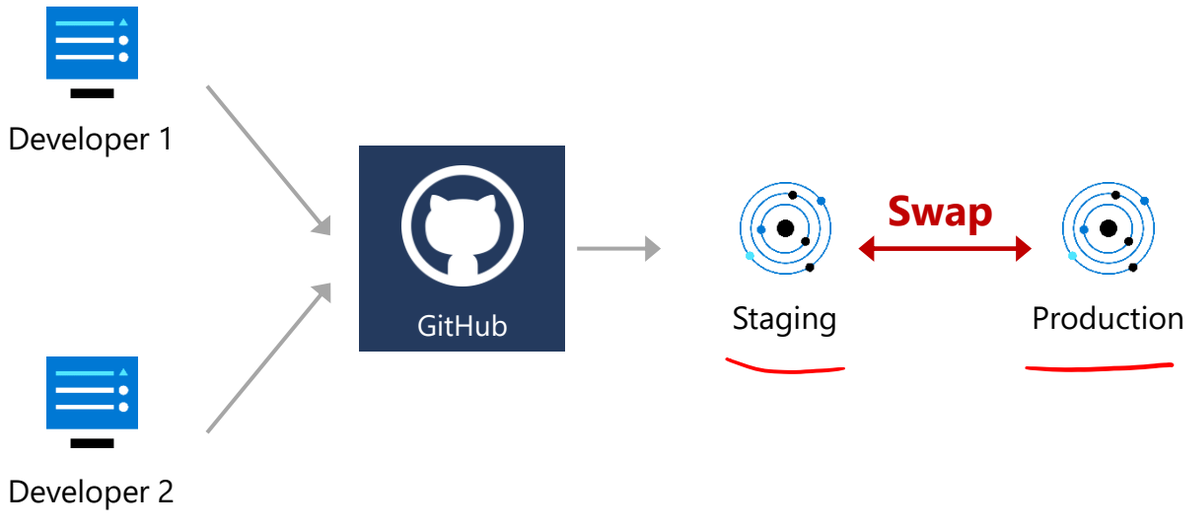
- Name must be unique
- Access using *azurewebsites.net* – can map to a custom domain
- Publish Code (Runtime Stack)
- Publish Docker Container
- Linux or Windows
- Region closest to your users
- App Service Plan



© Copyright Microsoft Corporation. All rights reserved.

Create Deployment Slots

Continuous Deployment with Stage Slot



Service Plan	Slots
Free, Shared, Basic	0
Standard	Up to 5
Premium	Up to 20
Isolated	Up to 20

Deploy to a different deployment slots (depends on service plan)

Validate changes before sending to production

Deployment slots are live apps with their own hostnames

Avoids a cold start – eliminates downtime

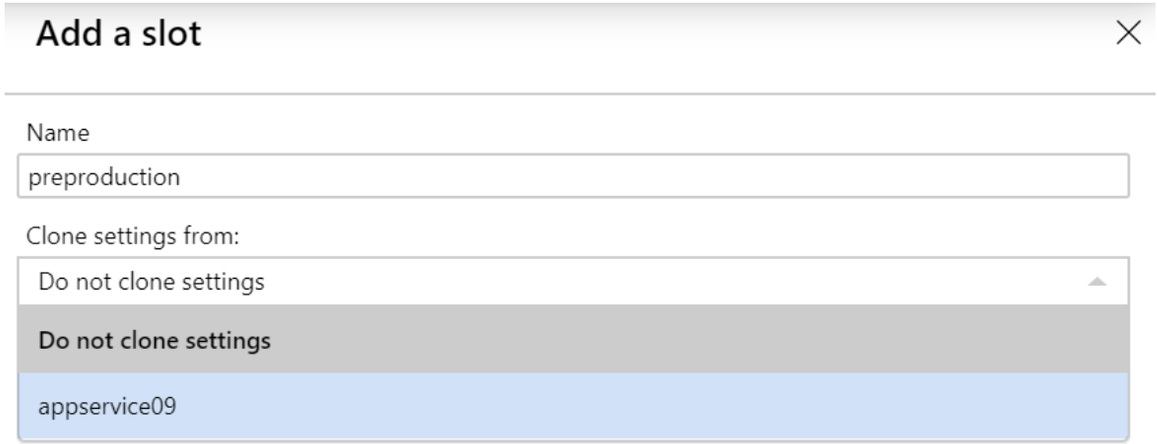
Fallback to a last known good site

Auto Swap when pre-swap validation is not needed

© Copyright Microsoft Corporation. All rights reserved.

Add Deployment Slots

- Select whether to clone an app configuration from another deployment slot
- When you clone, pay attention to the settings:
  - Slot-specific app settings and connection strings
  - Continuous deployment settings
  - App Service authentication settings
- Not all settings are sticky (endpoints, custom domain names, SSL certificates, scaling)
- Review and edit your settings before swapping



© Copyright Microsoft Corporation. All rights reserved.

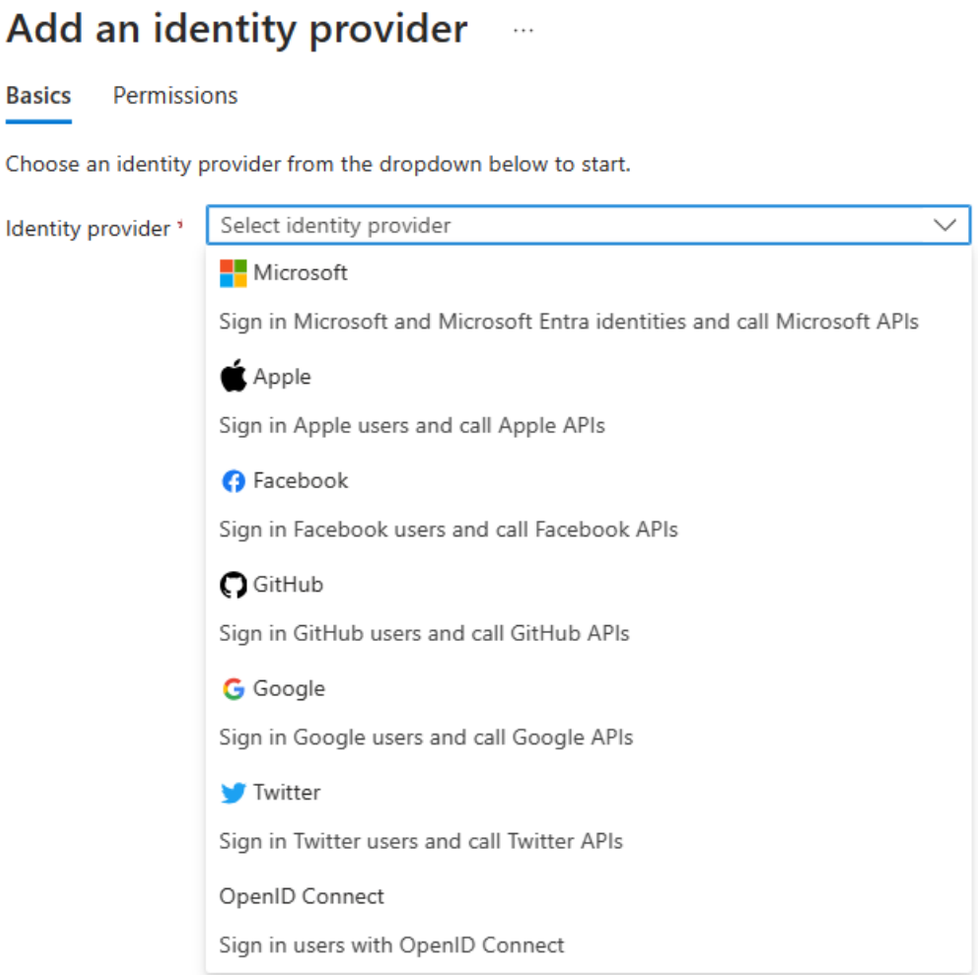
Secure an App Service

Authentication:

- Enable authentication – default anonymous
- Log in with a third-party identity provider

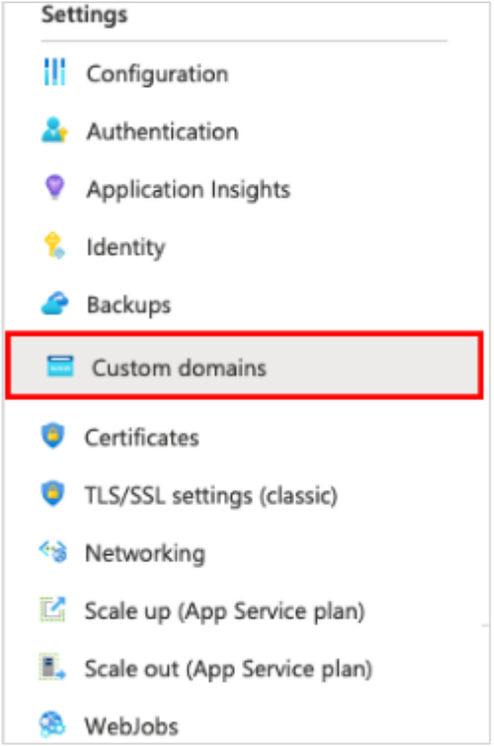
Security:

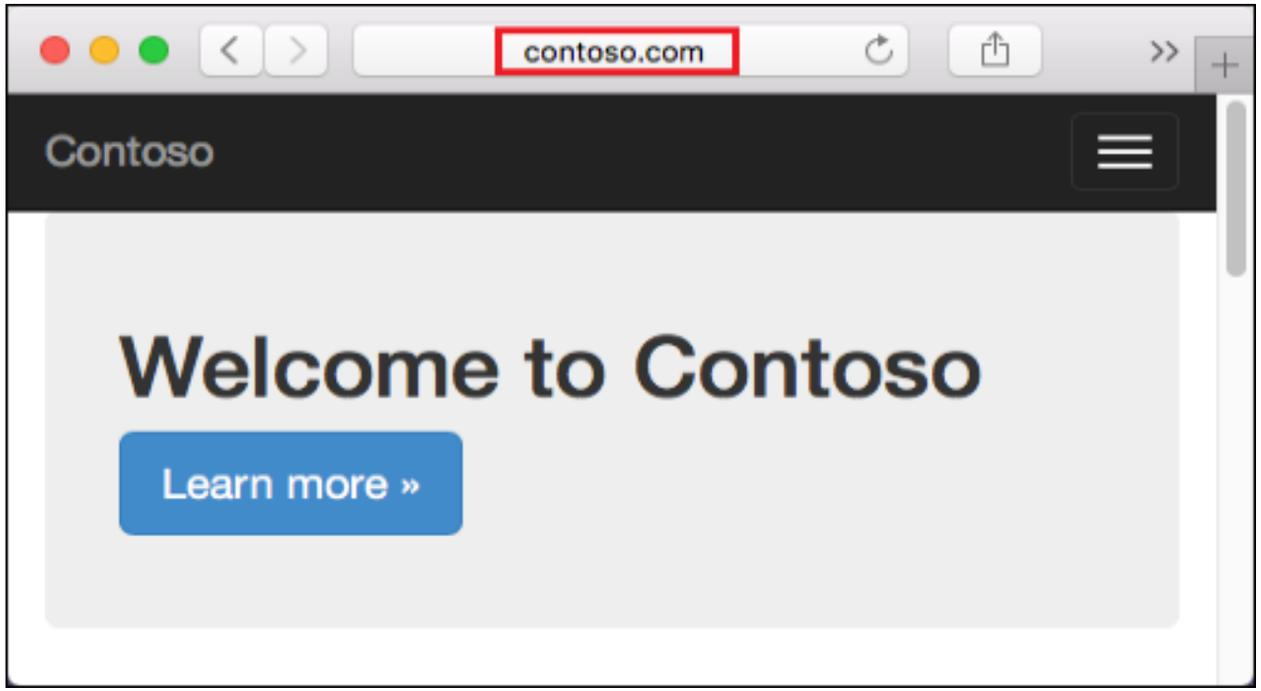
- Troubleshoot with Diagnostic Logs – failed requests, app logging
- Add an SSL certificate – HTTPS
- Define a priority ordered allow/deny list to control network access to the app
- Store secrets in the Azure Key Vault



© Copyright Microsoft Corporation. All rights reserved.

Create Custom Domain Names



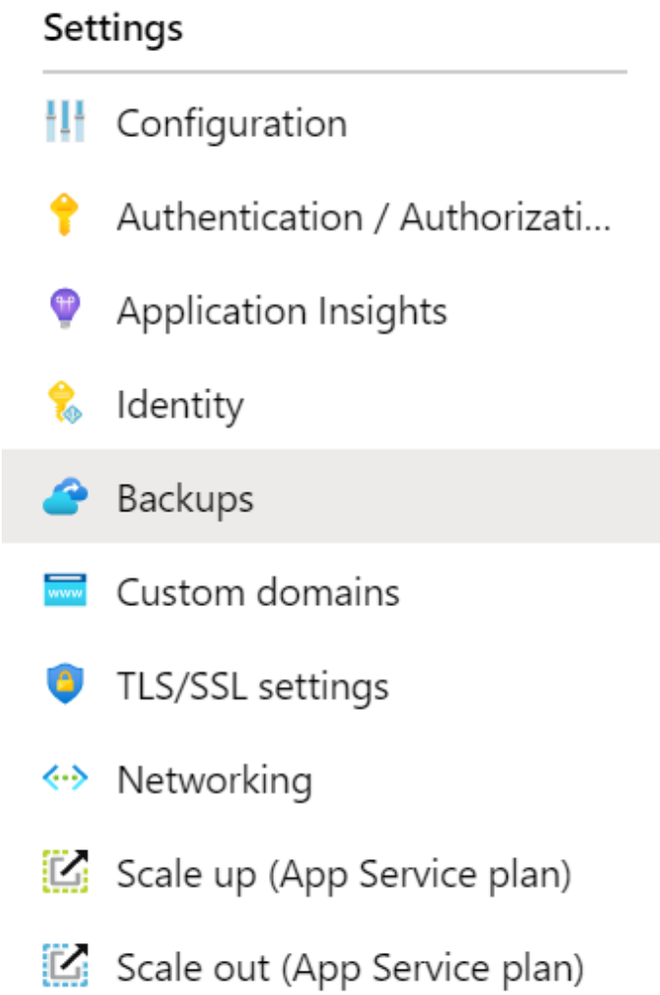


- Redirect the default web app URL
- Validate the custom domain in Azure
- Use the DNS registry for your domain provider – create a CNAME or A record with the mapping
- Ensure App Service plan supports custom domains

© Copyright Microsoft Corporation. All rights reserved.

Backup an App Service

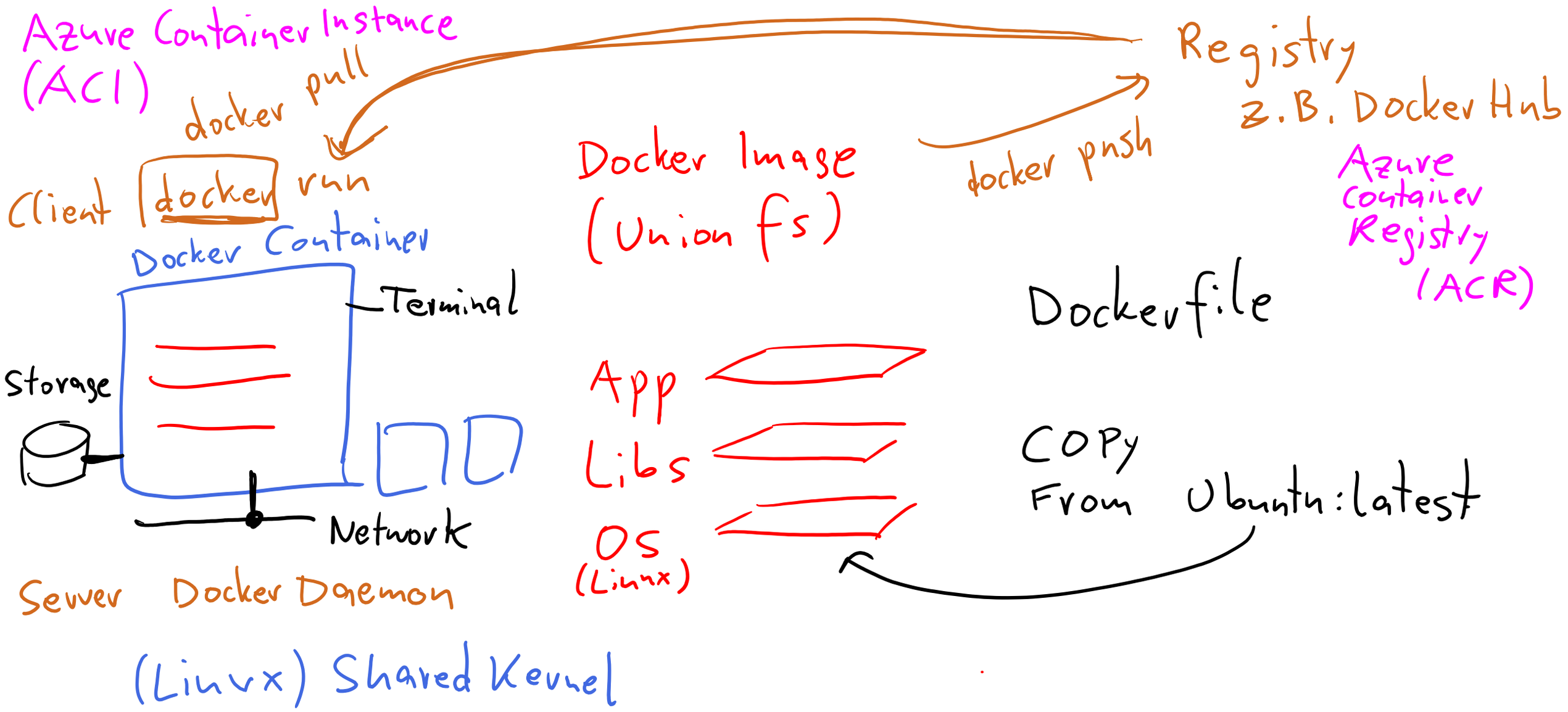
- Create app backups manually or on a schedule
- Backup the configuration, file content, and database connected to the app
- Requires Standard or Premium plan
- Backups can be up to 10 GB of app and database content
- Configure partial backups and exclude items from the backup
- Restore your app on-demand to a previous state, or create a new app



© Copyright Microsoft Corporation. All rights reserved.

Configure Azure Container Instances Apps

© Copyright Microsoft Corporation. All rights reserved.



© Copyright Microsoft Corporation. All rights reserved.

Learning Objectives - Configure Azure Container Instances

- Compare Containers to Virtual Machines
- Understand Container Images
- Review Azure Container Instances
- Implement Container Groups
- Demonstration – Configure Azure Container Instances
- Manage Containers with Azure Container Apps
- Compare container management solutions
- Demonstration – Configure Azure Container Apps
- Learning Recap

Exam objectives

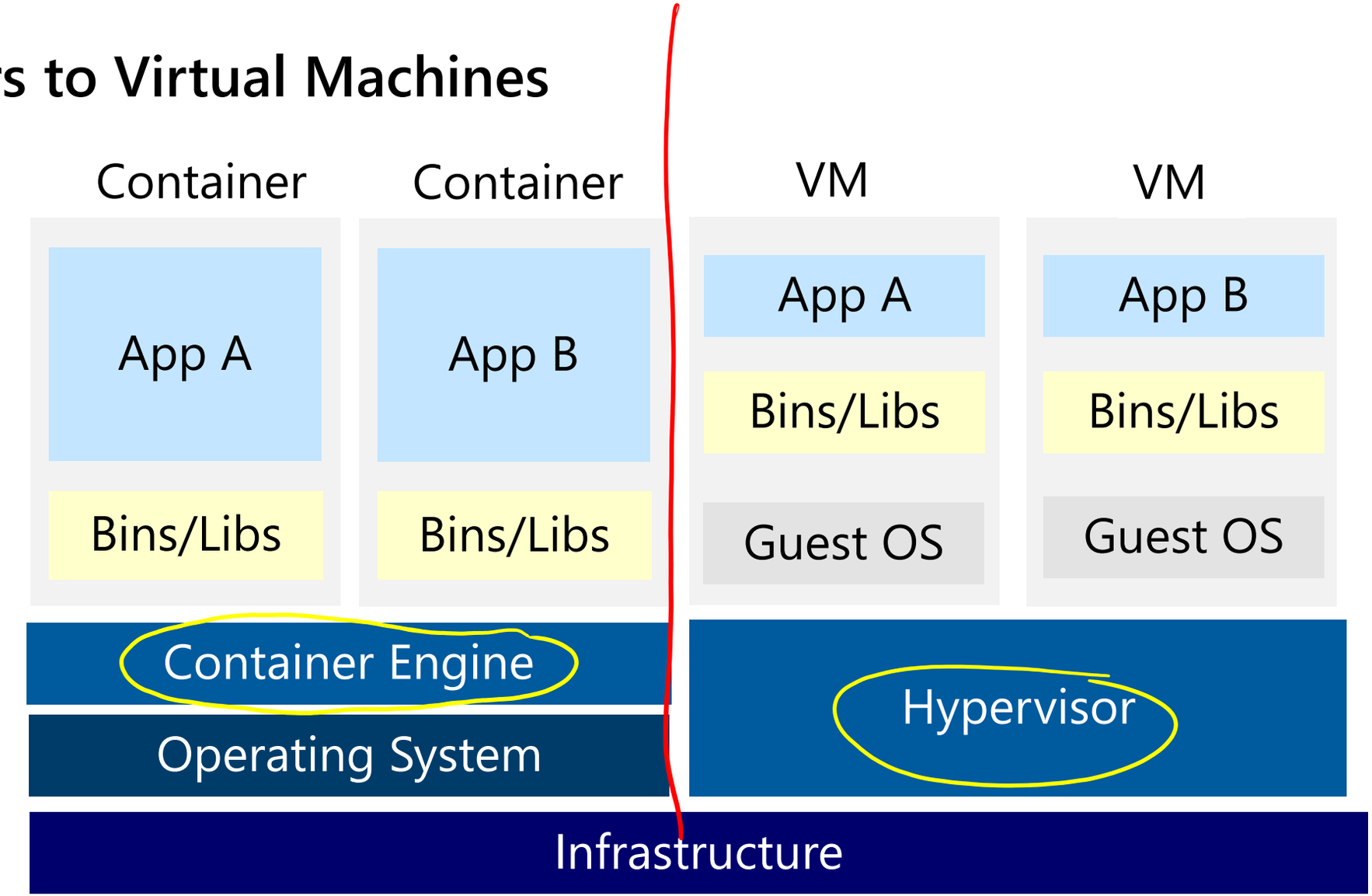
Provision and manage containers in Azure portal

- Create and manage an Azure container registry
- Provision a container by using Azure Container Instances (ACI)
- Provision a container by using Azure Container Apps (ACA)
- Manage sizing and scaling for containers, including ACI and ACA

© Copyright Microsoft Corporation. All rights reserved.

Compare Containers to Virtual Machines

- Isolation
- Operating System
- Deployment
- Persistent storage
- Fault tolerance



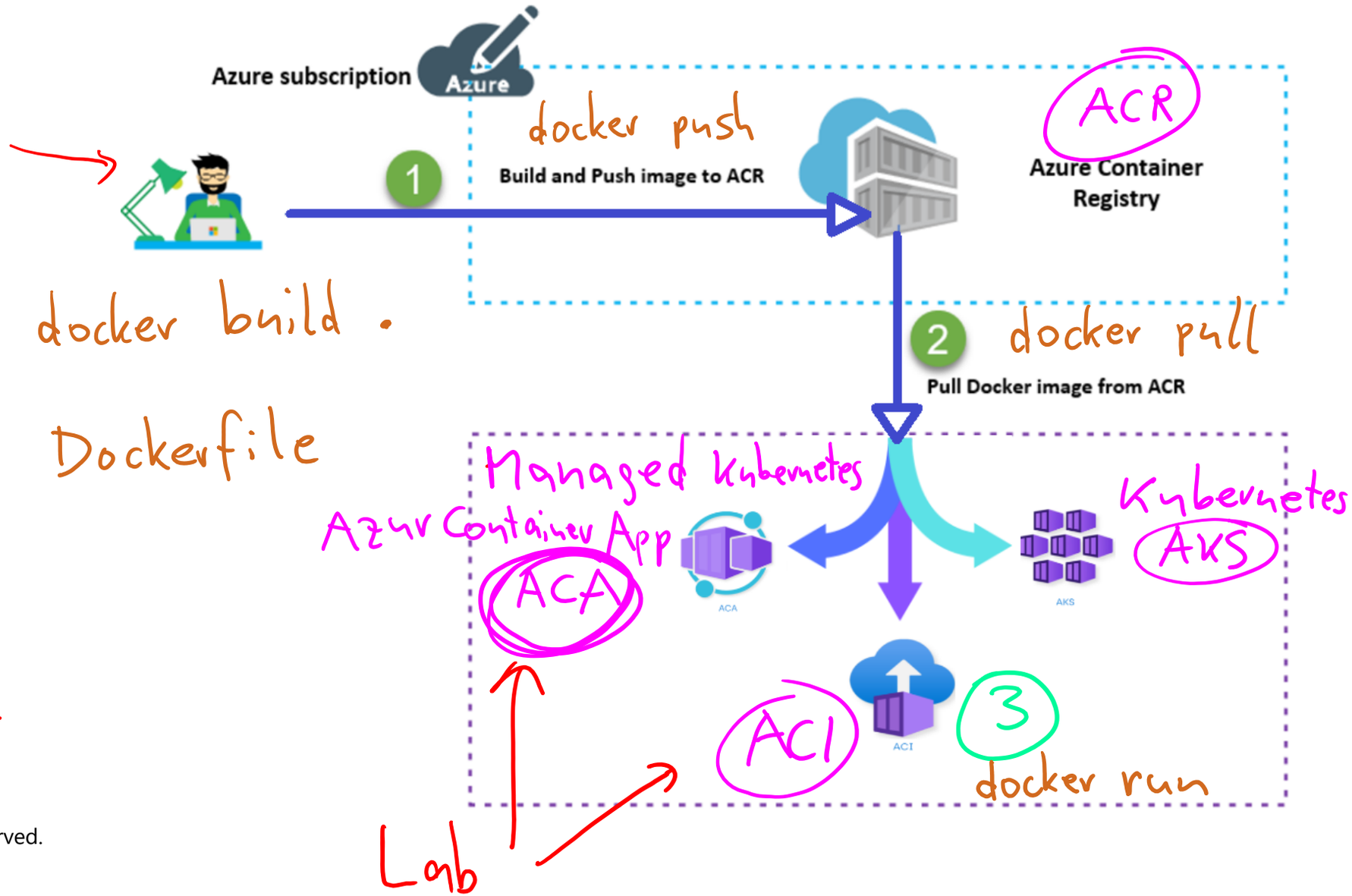
© Copyright Microsoft Corporation. All rights reserved.

Understand Container Images

Read Only

Supermario : v1
: latest

A container image is a lightweight, standalone, executable package of software that encapsulates everything needed to run an application.



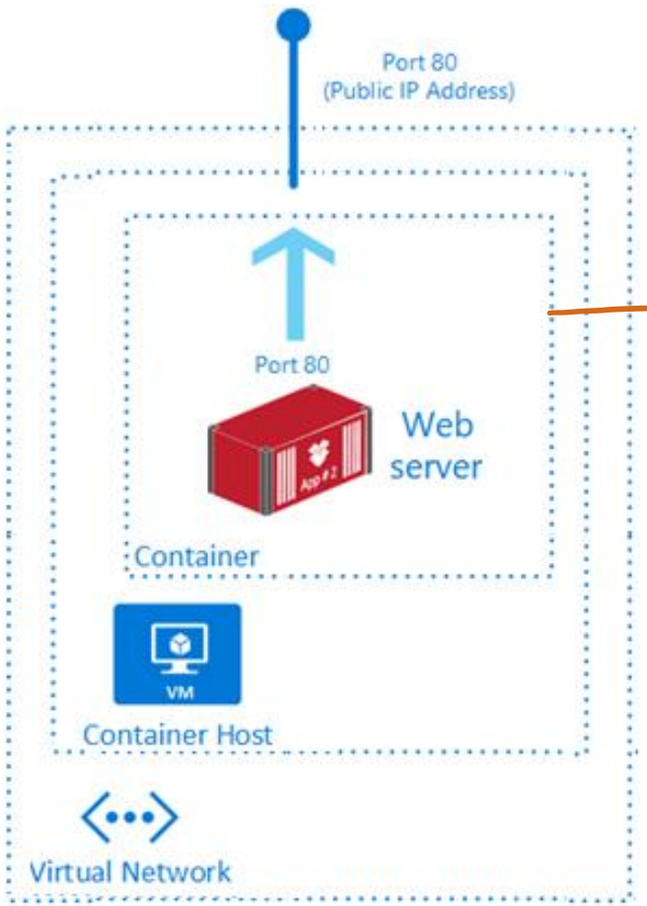
Apps → Stateless

© Copyright Microsoft Corporation. All rights reserved.

ACI

Review Azure Container Instances

- PaaS Service
- Fast startup times
- Public IP connectivity and DNS name
- Isolation features
- Custom sizes
- Persistent storage
- Linux and Windows Containers
- Co-scheduled Groups
- Virtual network Deployment



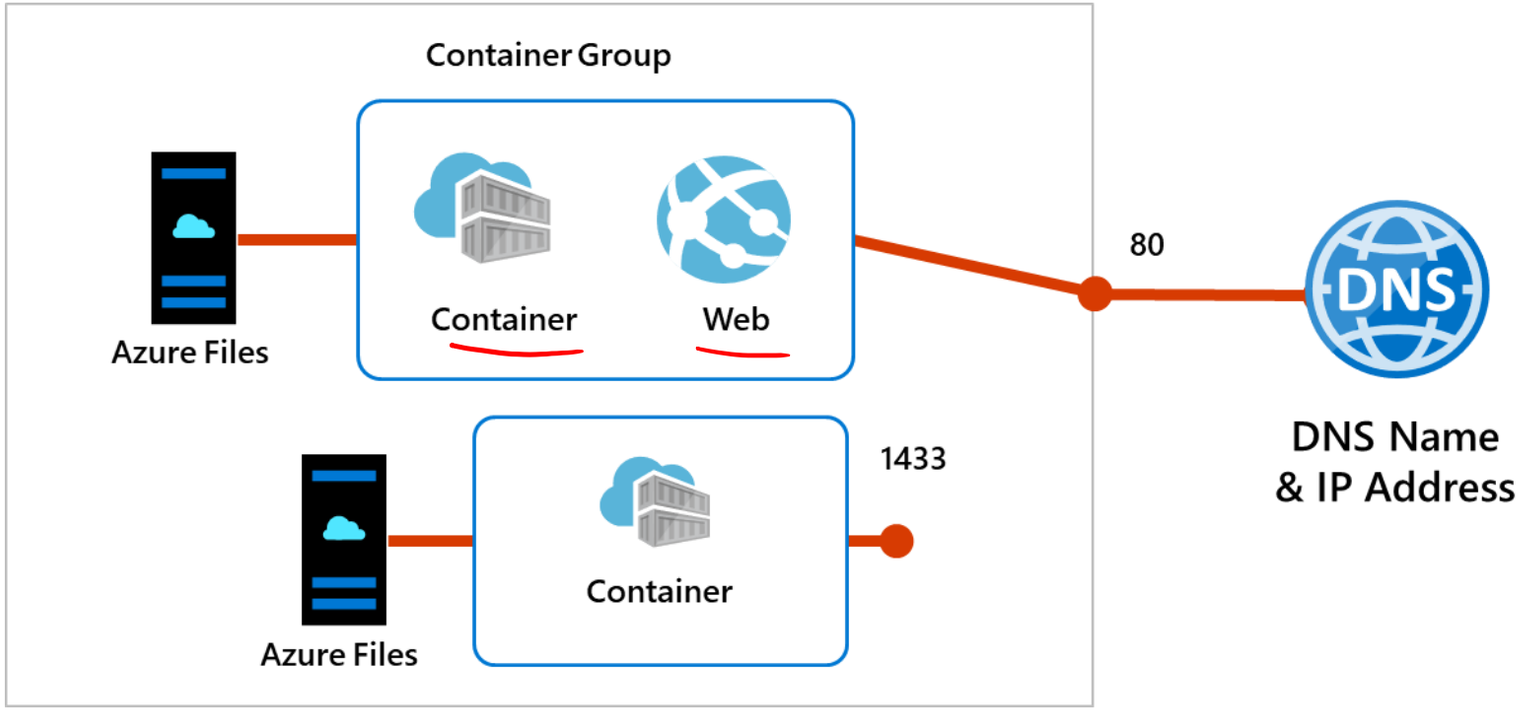
Terminal

Fastest way to run a container in Azure without provisioning a VM

© Copyright Microsoft Corporation. All rights reserved.

Implement Container Groups = Pod (kubernetes)

- Top-level resource in Azure Container Instances
- A collection of containers that get scheduled on the same host
- The containers in the group share a lifecycle, resources, local network, and storage volumes



© Copyright Microsoft Corporation. All rights reserved.

Cluster

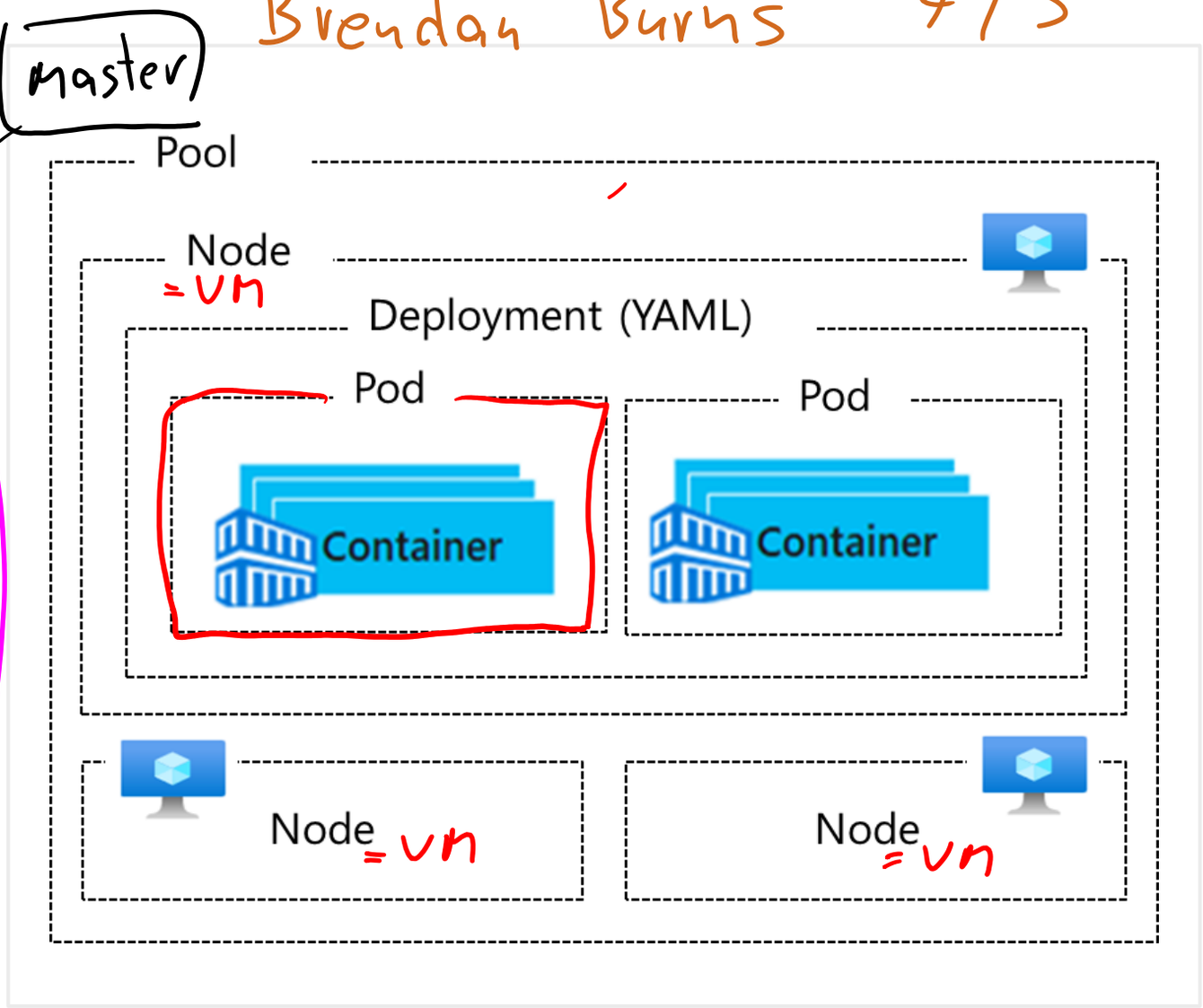
Kubernetes (Steuermann)

Brendan Burns 7/8

Understand AKS Terminology

Term	Description
Pools	Groups of nodes with identical configurations
Nodes	Individual VMs running containerized applications
Pods	Single instance of an application. A pod can contain multiple containers
Deployment	One or more identical pods managed by Kubernetes
Manifest	YAML file describing a deployment JSON

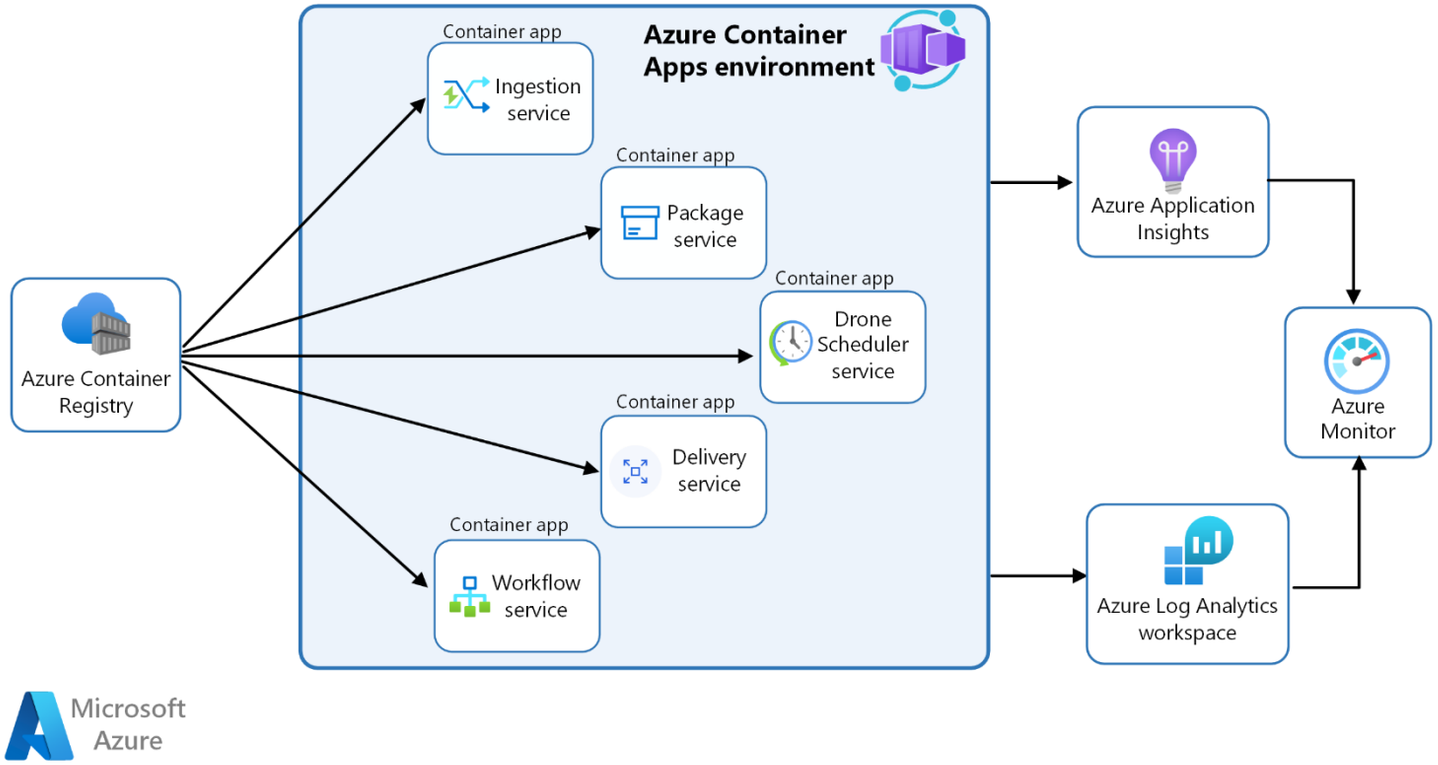
API



© Copyright Microsoft Corporation. All rights reserved.

Manage Containers with Azure Container Apps

- Alternative to Azure Kubernetes Service – manages container orchestration
- The Container App environment creates a secure boundary around the apps and jobs
- The Container App runtime manages the environment (OS upgrades, scaling, versioning, and failover)



© Copyright Microsoft Corporation. All rights reserved.

Compare container management solutions

	Azure Container Apps	Azure Kubernetes Service
Overview	Simplifies the deployment and management of microservices-based applications by abstracting away the underlying infrastructure.	Simplifies deploying a managed Kubernetes cluster in Azure by offloading the operational overhead to Azure.
Deployment	PaaS experience.	Offers more control and customization.
Management	Fully managed by Azure.	Partially managed by Azure (control plane).
Scalability	HTTP-based autoscaling and event-driven scaling.	Horizontal pod autoscaling and cluster autoscaling.
Use Cases	Rapid scaling and simplified management.	Complex, long-running applications that require full Kubernetes features.
Integration	Azure Logic Apps, Functions, and Event Grid for event-driven architecture.	Azure Policy for Kubernetes, Azure Monitor for containers, and Azure Defender for Kubernetes for comprehensive security and governance.

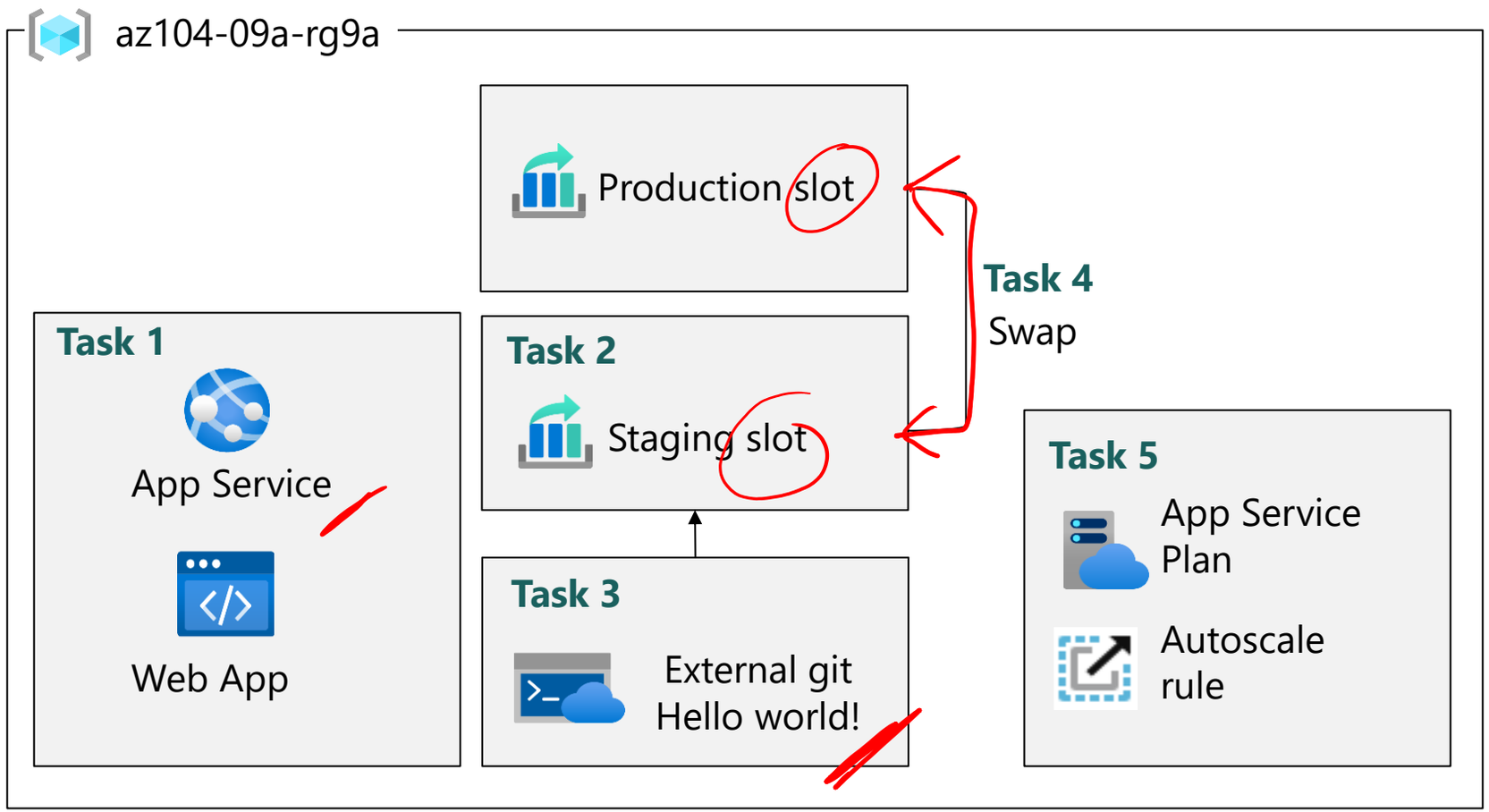
© Copyright Microsoft Corporation. All rights reserved.

Lab 09a – Implement Web Apps
Lab 09b – Implement Azure Container Instances
Lab 09c – Implement Azure Container Apps

© Copyright Microsoft Corporation. All rights reserved.

15

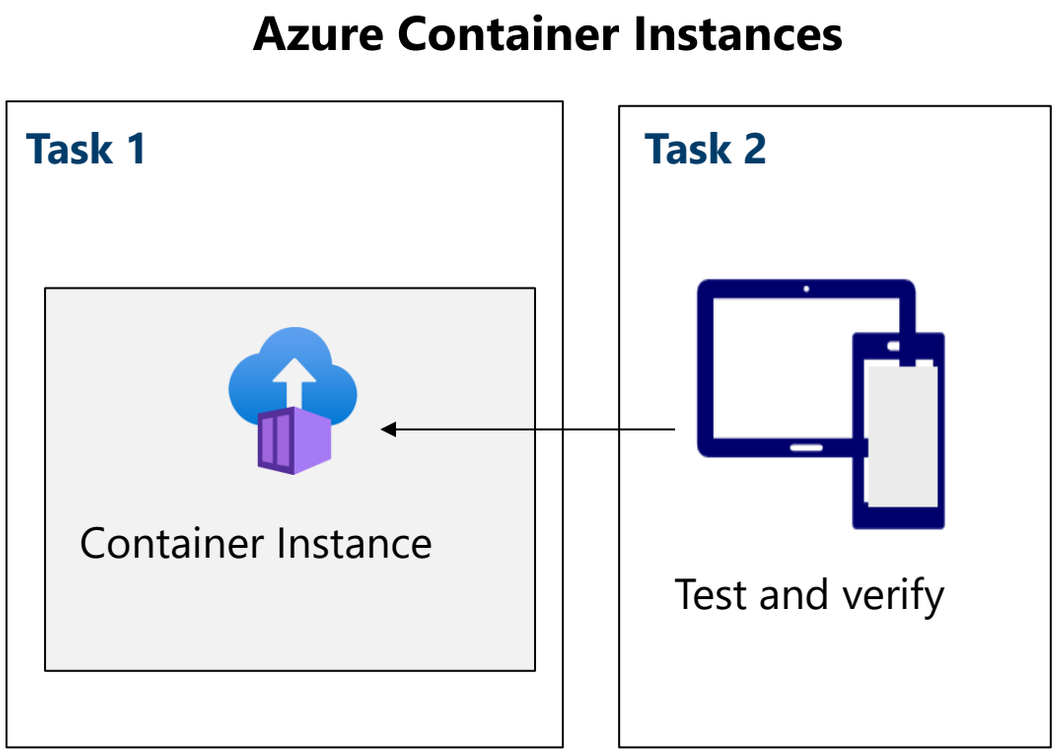
Lab 09a – Web App Architecture Diagram



Load Test

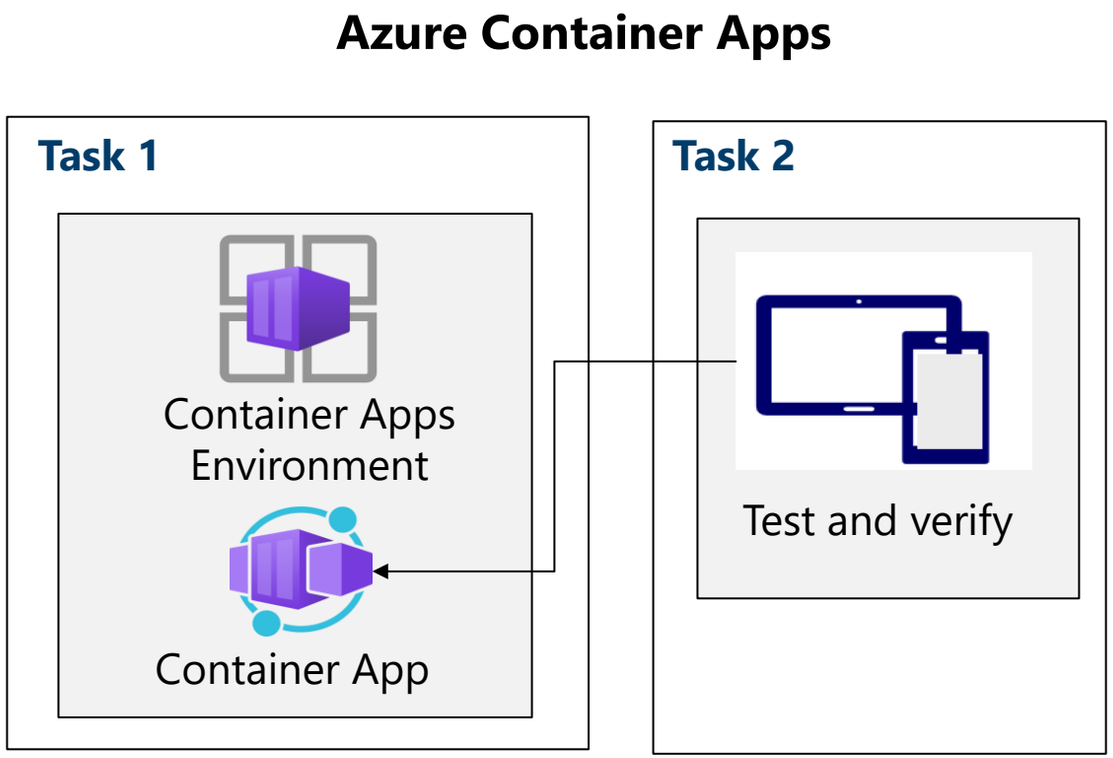
© Copyright Microsoft Corporation. All rights reserved.

Lab 09b – Azure Container Instances Diagram



© Copyright Microsoft Corporation. All rights reserved.

Lab 09c – Azure Container Architecture Diagram



© Copyright Microsoft Corporation. All rights reserved.

End of presentation

© Copyright Microsoft Corporation. All rights reserved.